THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: MIL-B-81820

THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.



3. SURFACE FINISH:

OUTER RACE: PLATING, WHEN SPECIFIED, SHALL BE CADMIUM PLATING PER QQ-P-416, TYPE II, CLASS 2.

PLATE ON THE OUTSIDE DIAMETER SURFACE AND ON THE FLAT BETWEEN THE OUTSIDE DIAMETER AND THE GROOVE.

PLATING RUNOUT MAY OCCUR EITHER IN THE GROOVE OR IN THE AREA BETWEEN THE GROOVE AND THE BALL.

THE PTFE LINER IN THE OUTER RACE INSIDE DIAMETER AND IN THE BALL BORE SHALL BE PROTECTED FROM EXPOSURE TO PLATING SOLUTIONS DURING PROCESSING.

BALL: PASSIVATE PER MIL-S-5002.

4. HARDNESS: BALL: Rc 43 MIN; OUTER RING: Rc 28 MIN, Rc 37 MAX BEFORE SWAGING.

5. NO-LOAD TORQUE: WHEN THE LETTER "K" IS PRESENT IN THE PART NUMBER LOWER VALUES OF NO-LOAD TORQUE ARE SPECIFIED PER TABLE I. IF THE MEASURED TORQUE OF A "K" TYPE BEARING IS LESS THAN 0.1 INCH-POUND, THE INTERNAL RADIAL CLEARANCE SHALL BE MEASURED AND SHALL NOT EXCEED THE VALUES IN TABLE II. THESE REQUIREMENTS APPLY TO THE TORQUE AND INTERNAL PLAY BETWEEN THE SPHERICAL BALL AND THE OUTER RING. THIS STANDARD DOES NOT DEFINE REQUIREMENTS FOR TORQUE OR INTERNAL PLAY BETWEEN THE BEARING BORE AND SHAFT.

TABLE II. INTERNAL CLEARANCE

| DASH NO. | MAXIMUM RADIAL PLAY | MAXIMUM AXIAL PLAY |
|---|---|---|
| -4K THRU -12K | 0.0007 INCH | 0.0028 INCH |
| -14K THRU -16K | 0.0010 INCH | 0.0040 INCH |

6. BREAK SHARP EDGES AND CORNERS AND REMOVE ALL BURRS AND SLIVERS.

7. DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES, DECIMALS ± .010, ANGLES ± 0.5.

NOTES:

1. WHEN TESTED TO THE FLUID CONTAMINATION AND SUB-ZERO TEMPERATURE REQUIREMENTS OF THE PROCUREMENT SPECIFICATION, THE OSCILLATING LOAD SHALL BE DECREASED TO 75% OF SPECIFIED LOAD.
2. WHEN FLUIDS AND ELEVATED TEMPERATURES (ABOVE 200°F) ARE BOTH PRESENT IN AN APPLICATION, THEN REDUCTIONS IN OPERATING LOADS OR BEARING LIFE MAY BE REQUIRED.
3. DASH NUMBER DESIGNATES NOMINAL BORE DIA IN SIXTEENTHS OF AN INCH.
4. DIMENSION "D" TO BE MET AFTER PLATING.
5. EXAMPLE OF PART NO.

M81820/1-6 P K = 3/8 NOMINAL BORE DIA, PH13-8 Mo BALL MATERIAL, CADMIUM PLATED, LOW TORQUE.

- ABSENCE OF LETTER INDICATES STANDARD BREAKAWAY TORQUE.
- LETTER "K" INDICATES LOW BREAKAWAY TORQUE (SEE TABLE I).
- ABSENCE OF LETTER INDICATES NO PLATING.
- LETTER "P" INDICATES CADMIUM PLATING (SEE REQUIREMENT 3).
- SIZE - BORE DIAMETER. (DASH NO. FROM TABLE I).
- MS NUMBER.

6. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.
7. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DoDISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

PREPARING ACTIVITY: NAVY — AS
CUSTODIANS: ARMY — AV NAVY —
AIR FORCE — 99 DLA —
REVIEW: DLA-IS
USER: ARMY-MI
PROJECT NUMBER: 3120-0692

**MILITARY SPECIFICATION SHEET**

TITLE
BEARING, PLAIN, SELF-ALIGNING, SELF-LUBRICATING, LINED BORE, LOW SPEED, NARROW, GROOVED OUTER RING, -65°F TO 325°F

SPECIFICATION SHEET NUMBER 31 JAN 90
MIL-B-81820/1

SUPERSEDING

AMSC — N/A FSC 3120

DISTRIBUTION STATEMENT
A. APPROVED FOR PUBLIC RELEASE; DISTRIBUTION IS UNLIMITED.



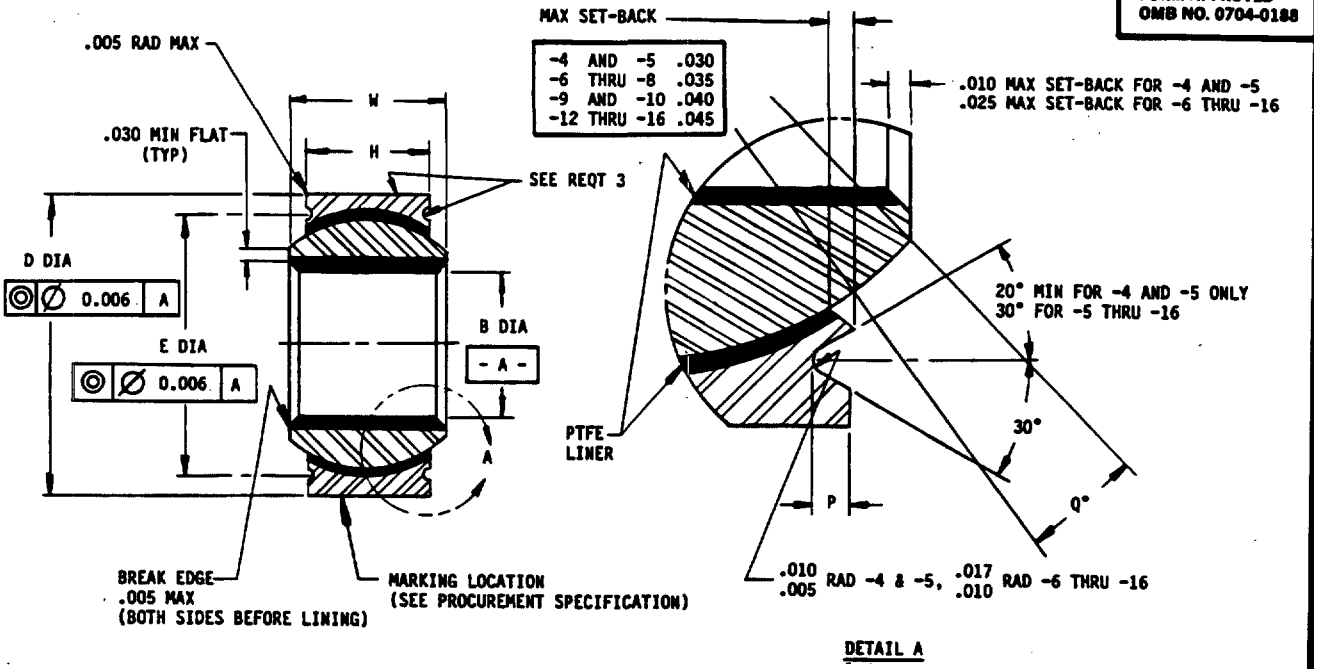


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THIS SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE.

FORM APPROVED
OMB NO. 0704-0188




DETAIL A

TABLE I. DIMENSIONS AND STRENGTHS

| DASH NO. | B +.0000 -.0010 | D +.0000 -.0005 | H ±.005 | P +.000 -.010 | Q MIN | W +.000 -.002 | E +.000 -.008 | STATIC LIMIT LOAD | | OSCILLATING LOAD LB | NO-LOAD ROTATIONAL BREAKAWAY TORQUE IN - LB | | WT LB MAX |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | RADIAL LB | AXIAL LB | | STANDARD | * "K" TYPE | |
| -4 | .2510 | .6562 | .250 | .025 | 10 | .343 | .594 | 5550 | 430 | 2650 | 1.0 - 5.0 | 0 - 0.5 | .020 |
| -5 | .3135 | .7500 | .281 | .035 | 10 | .375 | .660 | 7700 | 700 | 3700 | 1.0 - 15.0 | 0 - 1.0 | .030 |
| -6 | .3760 | .8125 | .312 | .035 | 9 | .406 | .712 | 10200 | 1100 | 4900 | 1.0 - 15.0 | 0 - 1.0 | .040 |
| -7 | .4385 | .9062 | .343 | .035 | 9 | .437 | .806 | 12950 | 1400 | 6700 | 1.0 - 15.0 | 0 - 1.0 | .050 |
| -8 | .5010 | 1.0000 | .390 | .055 | 8 | .500 | .876 | 17250 | 2100 | 8250 | 1.0 - 15.0 | 0 - 1.0 | .070 |
| -9 | .5635 | 1.0937 | .437 | .055 | 8 | .562 | .970 | 22150 | 3680 | 10600 | 1.0 - 15.0 | 0 - 1.0 | .090 |
| -10 | .6260 | 1.1875 | .500 | .055 | 8 | .625 | 1.063 | 27700 | 4720 | 13250 | 1.0 - 15.0 | 0 - 1.0 | .120 |
| -12 | .7510 | 1.4375 | .593 | .055 | 8 | .750 | 1.313 | 40600 | 6750 | 19400 | 1.0 - 15.0 | 0 - 1.0 | .210 |
| -14 | .8760 | 1.5625 | .703 | .055 | 8 | .875 | 1.438 | 55950 | 9350 | 26750 | 1.0 - 25.0 | 0 - 2.0 | .270 |
| -16 | 1.0010 | 1.7500 | .797 | .055 | 9 | 1.000 | 1.626 | 73800 | 12160 | 35250 | 1.0 - 25.0 | 0 - 2.0 | .390 |

* SEE REQUIREMENT 5 "NO-LOAD TORQUE" AND NOTE 5

REQUIREMENTS:

1. MATERIAL: BALL: PH13-8 Mo STEEL ALLOY PER AMS 5629, CONDITION H1000.
   OUTER RING: 17-4 PH STEEL ALLOY PER AMS 5643.
   LINER: POLYTETRAFLUOROETHYLENE (PTFE) SHALL BE INCLUDED IN THE LINER.

2. SURFACE TEXTURE: BALL DIA. Ra 8; BALL FACES Ra 16; OUTER RACE DIA Ra 32; ALL OTHER METALLIC SURFACES Ra 125. LINER SURFACES ARE EXEMPT FROM SURFACE TEXTURE MEASUREMENTS.

PREPARING ACTIVITY: NAVY — AS
CUSTODIANS: ARMY — AV NAVY —
AIR FORCE — 99 DLA —
REVIEW: DLA - IS
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